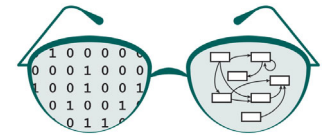


Extending the visualization capabilities of GRNsight: a web application for visualizing models of gene regulatory networks



Eileen J. Choe

Department of Electrical Engineering and Computer Science
Loyola Marymount University

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LMU|LA
Frank R. Seaver College
of Science and Engineering

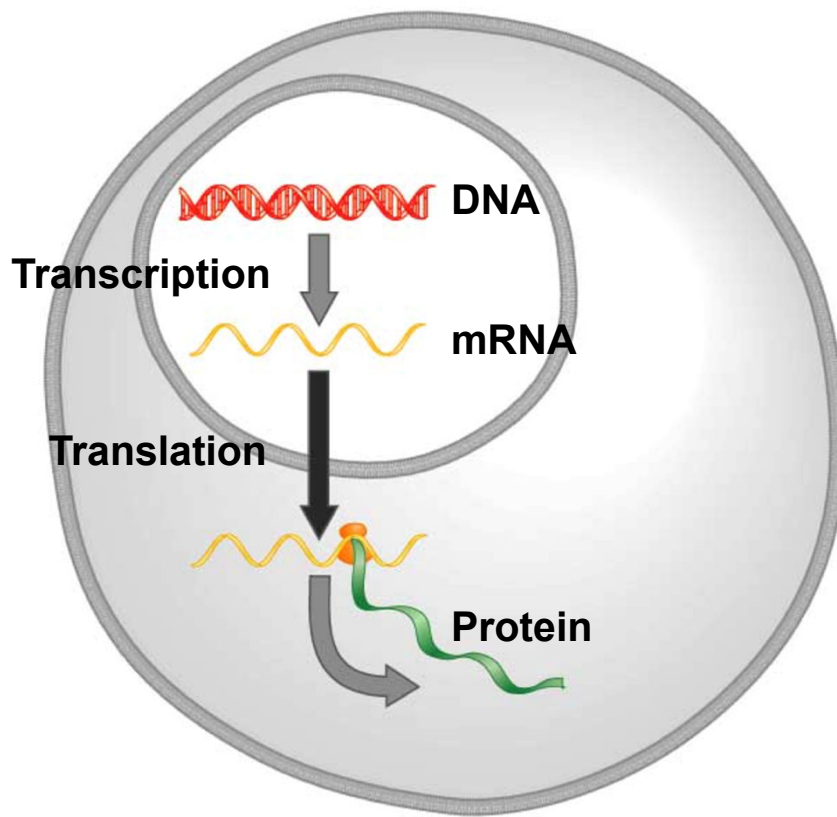
Outline

- **Gene regulatory networks (GRNs) consist of genes, transcription factors, and the regulatory relationships between them.**
- **GRNsight is an open source web application and service for visualizing models of gene regulatory networks.**
- **In GRNsight version 3.0, the Excel spreadsheet controller was redesigned to process data from the entire GRNmap input or output workbook.**
- **In GRNsight version 3.0, a node coloring feature for visualizing gene expression was added.**

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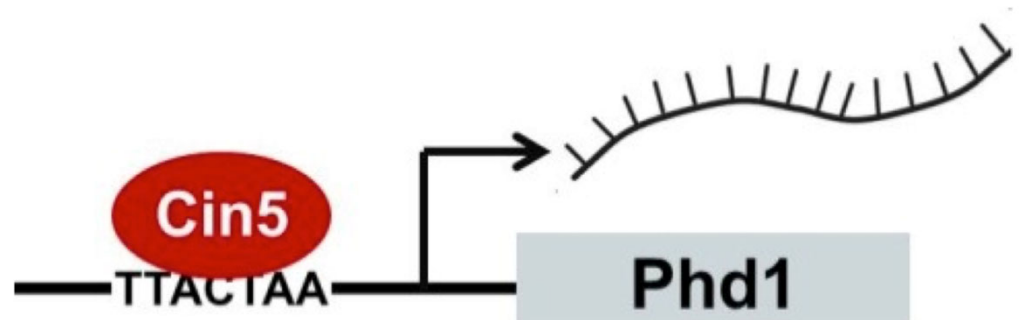
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A gene regulatory network (GRN) controls the level of expression of genes and proteins



Gene regulatory process

- Transcription factors **activate** or **repress** expression.
- To understand GRNs we want to quantify the **weight values** of these regulatory relationships



GRNmap models the dynamics of GRNs to estimate weight parameters (w)



- Magnitude of weights represent the influence of the transcription factor on its target gene
- **Positive** weights = **activation** of gene expression.
- **Negative** weights = **repression** of gene expression.

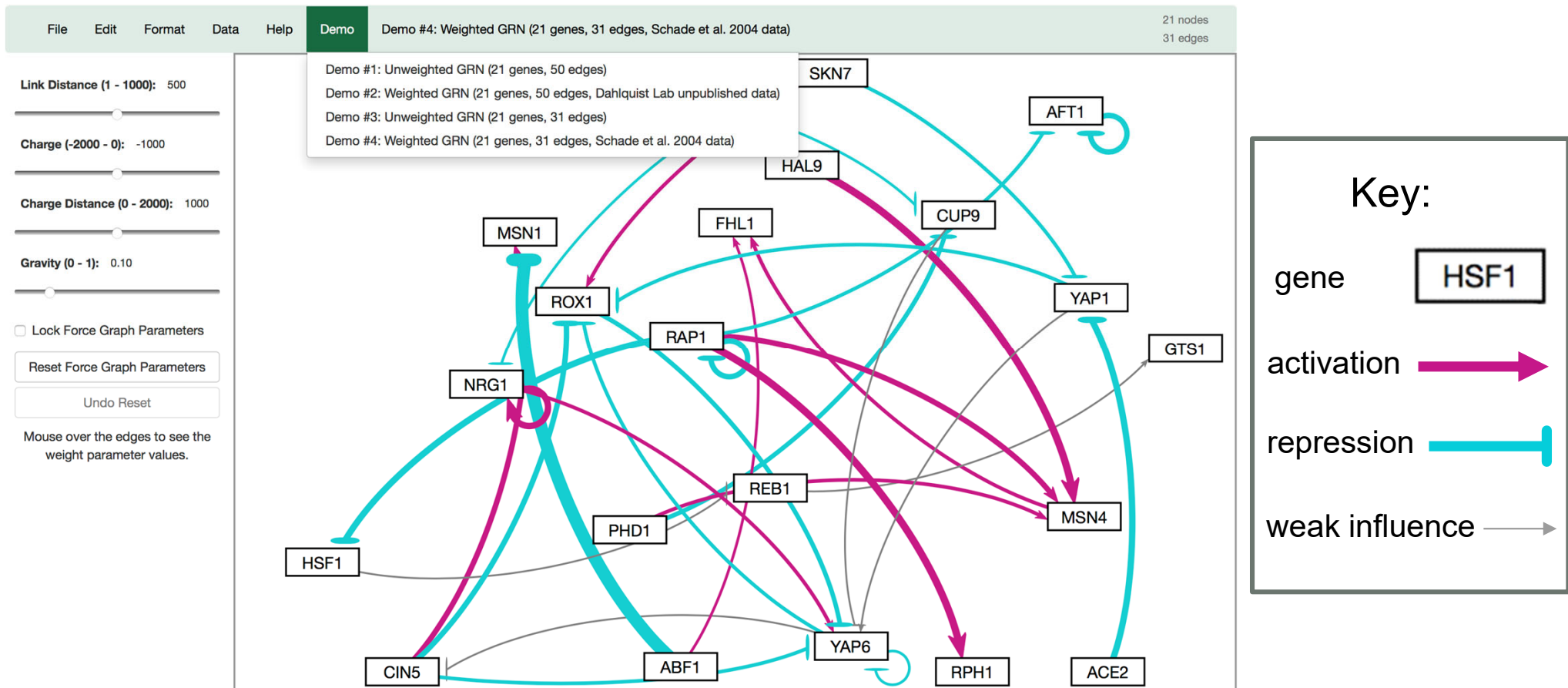
$$\frac{dx_i(t)}{dt} = \frac{P_i}{1 + \exp\left(-\left(\sum_j (w_{ij}x_j(t)) - b_i\right)\right)} - d_i x_i(t)$$

GRNmap differential equation

Outline

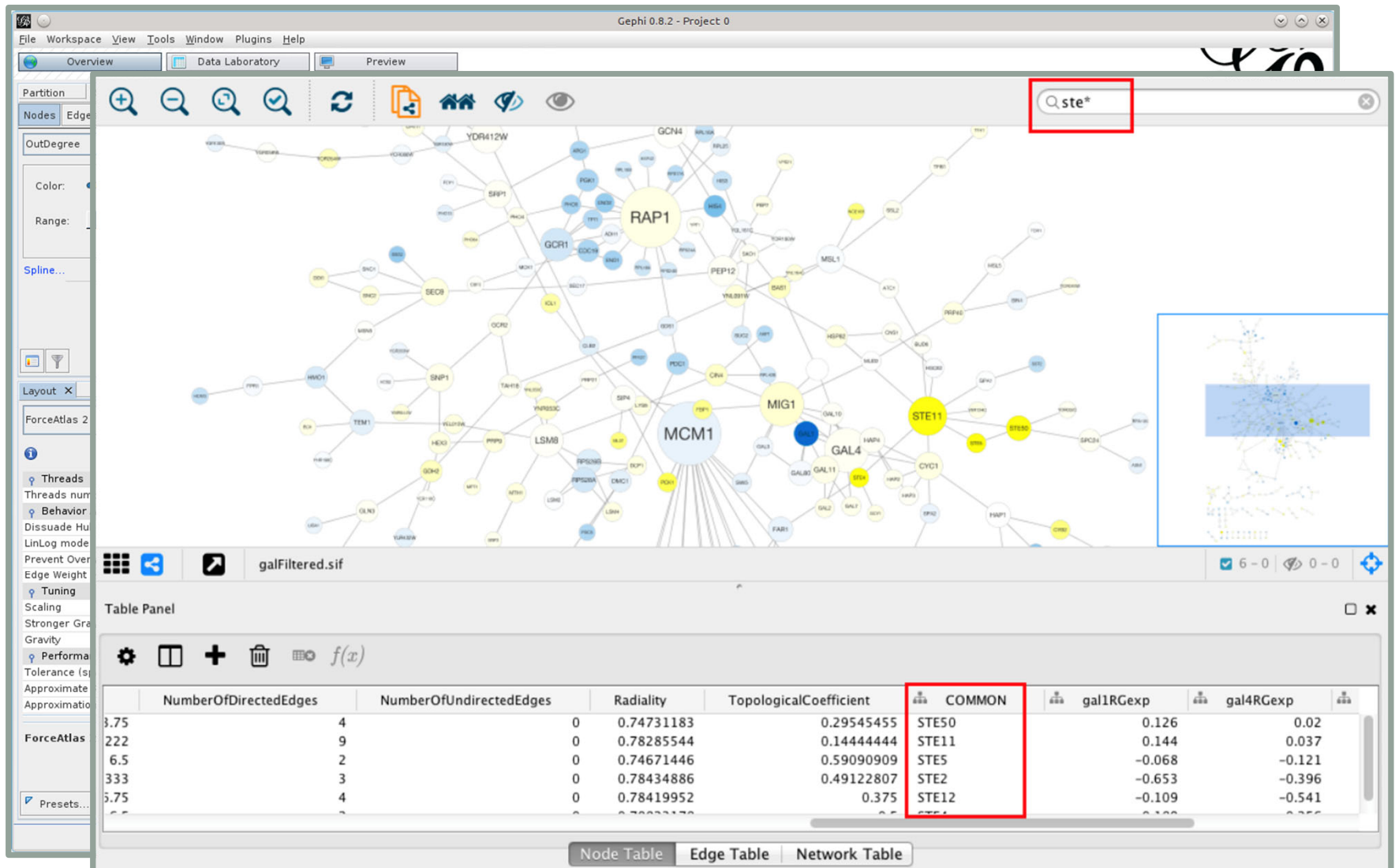
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GRNsight automatically produces visualizations of weighted networks from adjacency matrix inputs



Screenshot of GRNsight v1, released Aug 2016

Existing visualization software must be installed and is optimized for large-scale graphs



Cytoscape – cytoscape.org

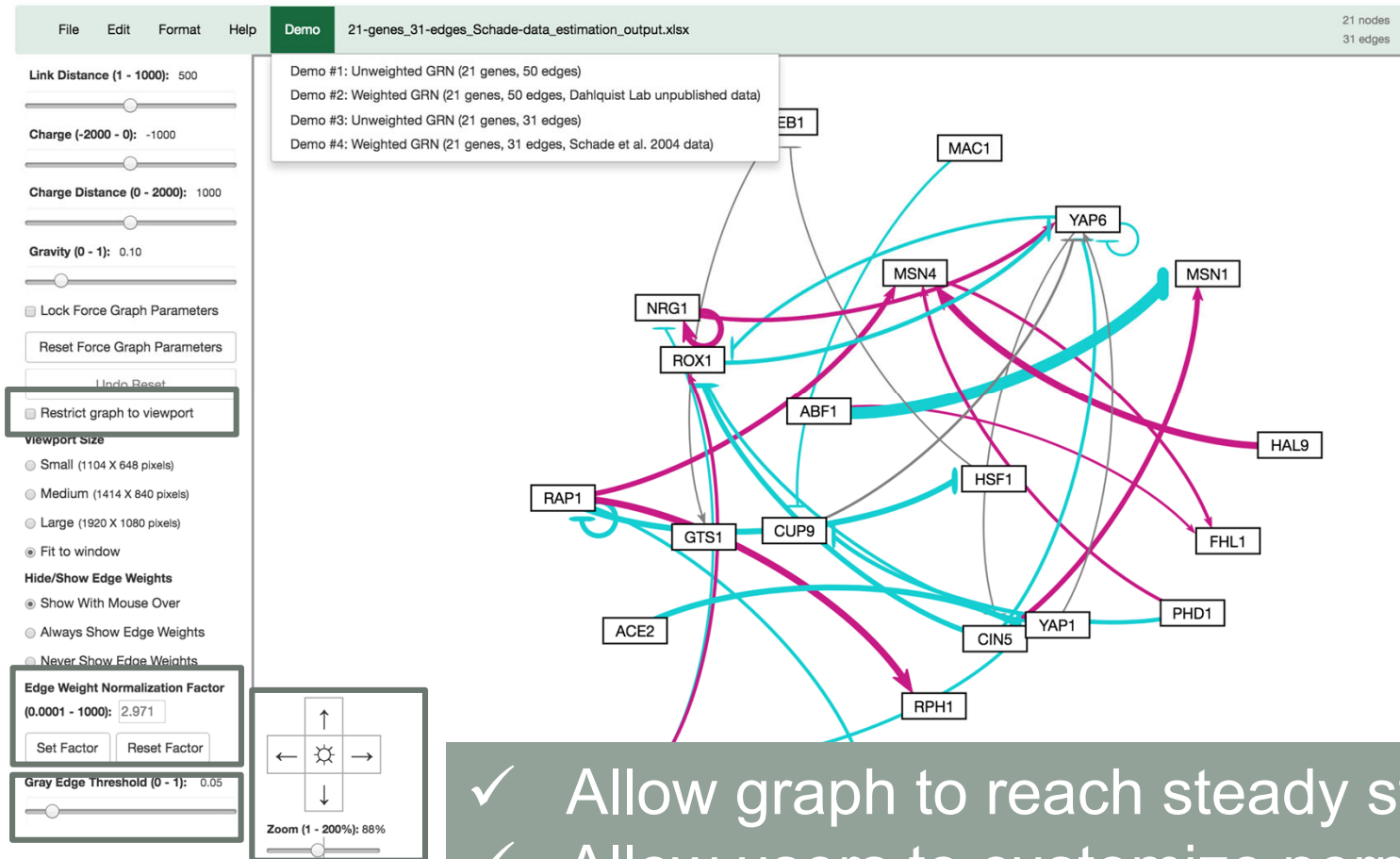
GRNsight fills a niche in bioinformatics visualization software

Cytoscape and Gephi	GRNsight
Optimized for large-scale graphs.	✓ Optimized for small- to medium-scale graphs.
Must be installed.	✓ Exists as a web application.
Several complex features in addition to visualization.	✓ Easy to use, and easily interpretable visualizations.

In GRNsight v2, we've expanded GRNsight's visual and analytic feature set

Beta v2.2.3

Disclaimer: This is the beta version of GRNsight for live testing of new functionality. For the most stable version, please go to the GRNsight home page.



**Screenshot of
GRNsight v2,
released June
2017**

- ✓ Allow graph to reach steady state
- ✓ Allow users to customize normalization factor for edge thickness
- ✓ Allow users to customize gray edge threshold

GRNsight v3 is focused on expanding visualization features

- What are alternative ways to visualize weighted networks other than the current force-directed graph?
 - Block Layout (Jen Shin, 2018)
- What are other datasets GRNsight can process and visualize?
 - GRNsight data structure redesign (Eileen Choe, 2018)
 - Node coloring (Eileen Choe, 2018)

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GRNsight was originally designed to process the “network” or “network_optimized_weights” sheet from a GRNmap input or output Excel workbook

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	cols regulators/rows targets	ABF1	ACE2	AFT1	CIN5	CUP9	FHL1	GTS1	HAL9	HSF1	MAC1	MSN1	MSN4	NRG1	PHD1	RAP1	REB1	ROX1	RPH1	SKN7	YAP1	YAP6
2	ABF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	ACE2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	AFT1	0	0	-0.8966	0	0	0	0	0	0	0	0	0	0	0	-0.4030	0	0	0	0	0	0
5	CIN5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.0450
6	CUP9	0	0	0	0	0	0	0	0	0	-0.1882	0	0	0	-0.6510	0	0	0	0	0	0	0
7	FHL1	0.1562	0	0	0	0	0	0	0	0	0	0	0.6121	0	0	0	0	0	0	0	0	0
8	GTS1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0778	0	0	0	0	0
9	HAL9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	HSF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1.2321	0	0	0	0	0	0
11	MAC1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	MSN1	-2.9707	0	0	0.9393	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	MSN4	0	0	0	0	0	0	0	1.4283	0	0	0	0	0	0.5447	1.0131	0	0	0	0	0	0
14	NRG1	0	0	0	0	0	0	0	0	0	0	0	0	1.2341	0	0	0	0	0	-0.1852	0	0
15	PHD1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	RAP1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.8890	0	0	0	0	0	0
17	REB1	0	0	0	0	0	0	0	0	-0.0102	0	0	0	0	0	0	0	0	0	0	0	0
18	ROX1	0	0	0	-0.9278	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.5744	-0.4315	-0.5071
19	RPH1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.4999	0	0	0	0	0	0
20	SKN7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
21	YAP1	0	-1.3615	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.4082	0	0
22	YAP6	0	0	0	-0.5312	-0.1293	0	0	0	0	0	0	0	0.6215	0	0	0	-0.7503	0	0	0.0146	-0.3027

Adjacency matrix values correspond to the magnitude of a regulatory relationship of a regulator to a target

GRNmap workbooks have other worksheets that contain additional data about the GRN

3 structural categories of GRNmap Excel sheets:

• 2-column worksheets

- Simple target – value structure for each gene
- e.g. “production_rates”, “degradation_rates”

id	production_rate
ACE2	0.2236
ASH1	0.4332
CIN5	0.2009
GCR2	0.1925
GLN3	0.3224
HAP4	0.2718
HMO1	0.0990

• expression

- Time Series data
- e.g. “wt_log2_expression”, “wt_log2_optimized_espression”

id	15	15	15	15	30	30	30	30
ACE2	-0.1718	-0.2792	-0.1752	-0.6884	-0.2436	0.1581	-0.4253	
ASH1	-1.3229	-0.4236	-1.8945	-1.6929	-2.2704	-1.0939	-1.3098	-1.017
CIN5	-0.6685		-0.4322	-0.1191	-0.383	0.2693	-0.0757	0.5975
GCR2	-0.3155	1.4649	-0.8029	0.5141	-0.7654	-1.1456	0.5096	-0.2957
GLN3	0.9621	-0.3518	0.3101	-0.1589	0.5869	0.0341	0.486	0.4342
HAP4	0.0633	0.1712	-1.9531	-0.2151	-1.463	0.3025	-2.2646	1.4417
HMO1	-0.9169	1.2958	0.427	-0.0712	1.5911	1.2566	0.391	-0.3126
MSN2		-0.5489	0.2491	0.9036	-0.5302	0.016	-0.0997	1.0787
SFP1	1.0932	0.4726	1.1255	0.6263	0.5256	-0.3325	1.6525	-0.1135
STB5	1.259	1.0717	0.916	0.804	-1.9025	-0.4406	0.5659	1.258
SWI4	-0.4763	-0.4529	-0.3336	-0.5489	2.5266	0.4041	0.0257	-0.2142
SWI5	-0.1331	0.1802	-0.3214	-0.0182	0.8526	-0.4332	0.2249	-0.7035
YHP1	-1.2421	0.3344	-0.735	0.2935	-1.2684	-0.8782	1.0021	-0.7668
YOX1	-1.3434	-0.9354	-1.7084	-0.2822	-0.0589	-0.7158	-0.1155	-2.4802
ZAP1	0.8344	0.8398	0.9184	0.4747	2.3845	0.1525	0.6511	-0.8263

• metadata

- Data about GRNmap simulation parameters
- e.g. “optimization_parameters”

optimization_parameter	value					
alpha	2.00E-03					
kk_max	1					
MaxIter	1.00E+08					
TolFun	1.00E-06					
MaxFunEval	1.00E+08					
TolX	1.00E-06					
production_function	Sigmoid					
L_curve	0					
estimate_params	1					
make_graphs	1					
fix_P	0					
fix_b	0					
expression_timepoints	15	30	60			
Strain	wt	dcin5	dglN3	dhap4	dhmo1	dzap1
simulation_timepoints	0	5	10	15	20	25

GRNsight's Excel spreadsheet parser was redesigned to process the entire GRNmap output

1. Parse, or analyze, the syntax (structure) of data and break it down into logical components
2. Transform data from Excel spreadsheets into JSON
 - JSON (JavaScript Object Notation) is a lightweight data format optimal for use in data visualization

id	production_rate
ACE2	0.2236
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HMO1	0.0990



```
production_rates: {  
  ACE2: 0.2236,  
  ASH1: 0.4332,  
  CIN5: 0.2009,  
  ...  
}
```

- Complete internal JSON representation of Excel input is first step towards expanding visualization capabilities of GRNsight

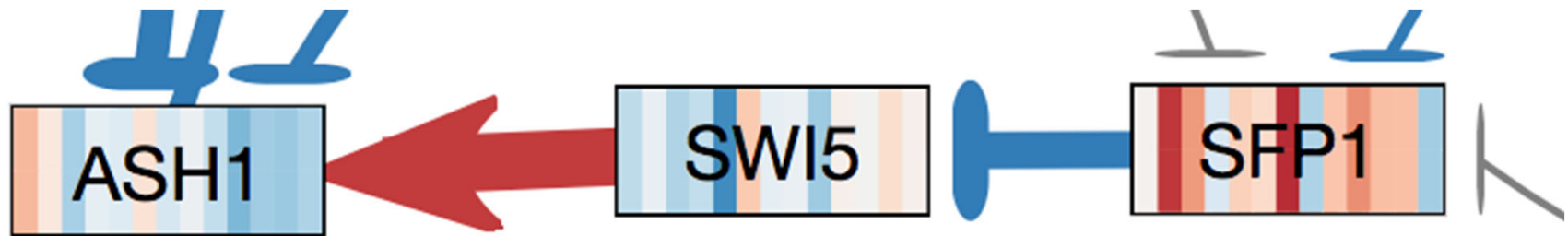
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Microarray experiments capture gene expression levels at specified time points

- background info + diagram
- // TODO work on this slide
- Gene expression data was one type of data added to the GRNsight internal JSON data structure in v3
- Node coloring framework developed to increase the density of information visualized by GRNsight, allowing users to learn more about data through visual inspection

Gene expression is visualized in GRNsight by coloring the nodes with a heat map corresponding to expression levels over time



- Rectangular nodes are horizontally subdivided by the number of data time points
- The color of each horizontal “slice” corresponds to the expression value of that gene at the particular time point
 - Red = activation (+ expression)
 - Blue = repression (- expression)

A node coloring menu allows users to modify parameters of node coloring

Node Coloring Menu

Top Dataset

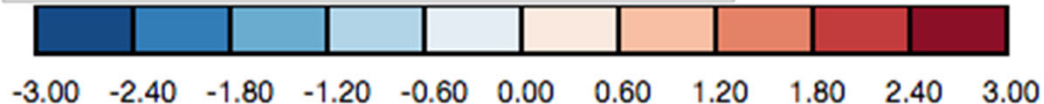
wt_log2_expression

Bottom Dataset

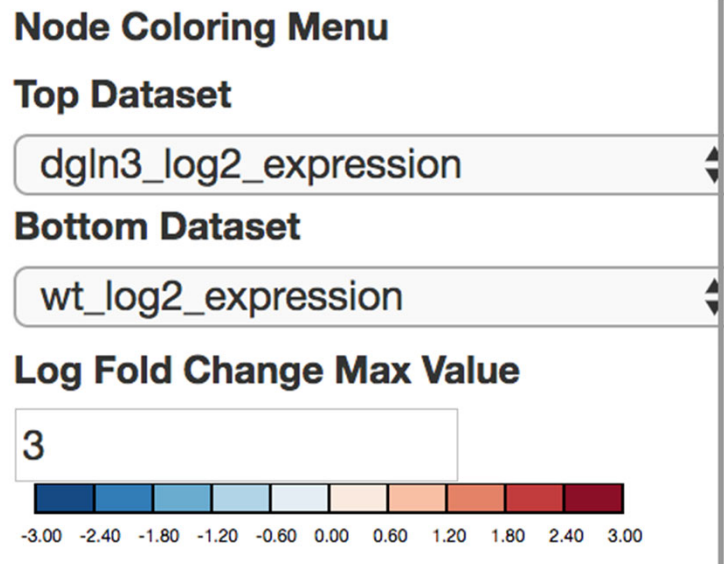
wt_log2_expression

Log Fold Change Max Value

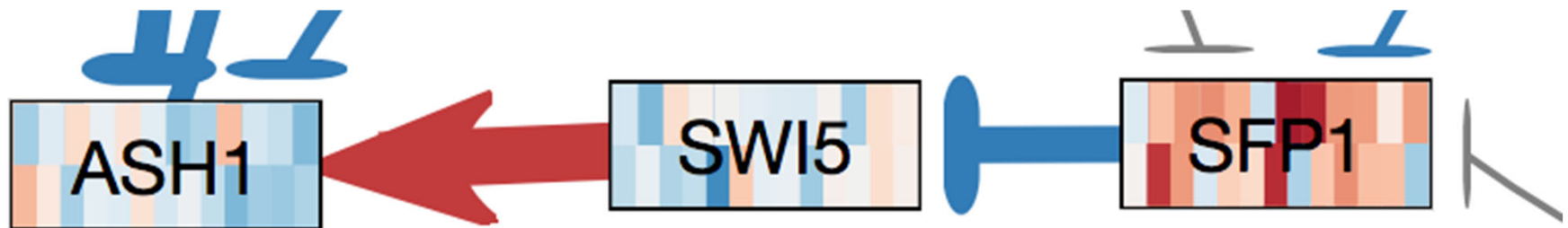
3



Users can manually select the dataset(s) to use for node coloring



- Options are auto generated by automatically detecting the expression sheets in the Excel workbook
- Two different datasets can be selected to visually compare expression data



Users may override the default log fold change max value

Node Coloring Menu

Top Dataset

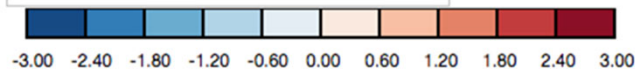
dcin5_log2_expression

Bottom Dataset

dcin5_log2_expression

Log Fold Change Max Value

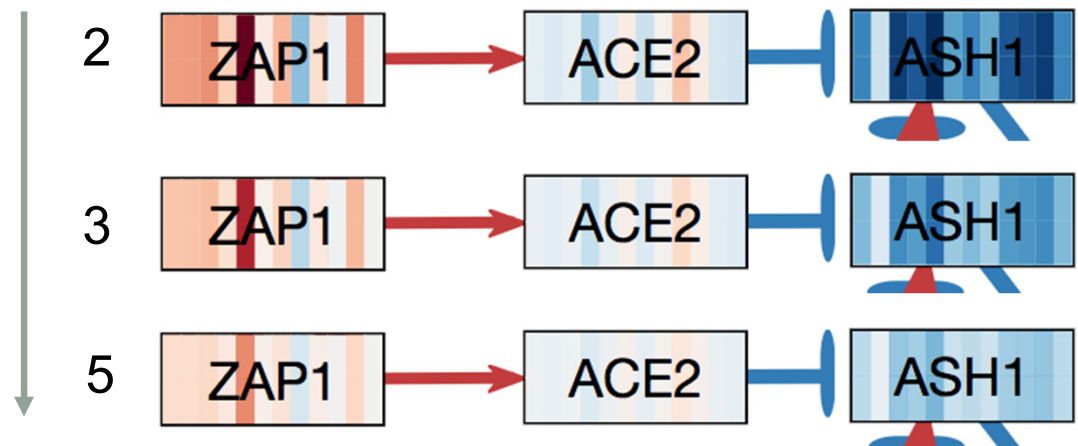
3



- Log fold change max value determines how expression values “map” to the coloring legend
- Default value arbitrarily set to 3

Users may change value to increase/decrease color intensity of their visualization

Log Fold Change Max Value



Visual inspection of gene expression is crucial to verifying the correctness of the GRNmap network simulation

- For example...

Conclusion

- **Gene regulatory networks (GRNs) consist of genes, transcription factors, and the regulatory relationships between them.**
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GRNsight application and code are available under the open source BSD license



<http://dondi.github.io/GRNsight/>



<https://github.com/dondi/GRNsight>

Current and future work documented via Github Issues

- User interface menu redesign
- Additional options for node coloring
 - User selected color scheme
 - Animation

Acknowledgements



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Dr. Kam D. Dahlquist

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GRNsight Team

Mihir Samdarshi

Jen Shin

Eileen Choe

GRNmap Team

Eddie Azinge

Maggie O'Neil

Brandon Klein

Justin Kyle Torres

Wet Lab Team

Nika Vafadari

Katherine Scheker

Nick Collins